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CS 2300 : Discrete Computational Structures
Summer Semester, 2025
ASSIGNMENT #5
Due Date: Friday, June 27

Suggested Reading: Rosen Section 2.5; LLM Chapter 8.1

For the problems below, explain your answers and show your reasoning.

Set Cardinality

1. [12 Pts] Show that the following sets are countably infinite, by first giving an enumeration, and then defining a bijection between $\mathbb{N}$ and that set. You do not need to prove that your function is bijective.

- (a) [6 Pts] the set of integers divisible by 5

Enumerate:

The set of integers divisible by 5 is $\{\dots, -15, -10, -5, 0, 5, 10, 15, \dots\}$

A possible enumeration is $0, 5, -5, 10, -10, 15, -15, 20, -20, \dots$

This alternates between positive and negative multiples of 5, including 0.

Define Bijection:

Define: $f : \mathbb{N} \rightarrow \mathbb{Z}_5$, where $\mathbb{Z}_5$ is the set of integers divisible by 5:

$$f(n) = \begin{cases} 5 \cdot \left(\frac{n}{2}\right) & \text{if } n \text{ is even} \\ -5 \cdot \left(\frac{n+1}{2}\right) & \text{if } n \text{ is odd} \end{cases}$$

I have:

An enumeration of the set: $0, 5, -5, 10, -10, \dots$ and a bijection $f : \mathbb{N} \rightarrow \mathbb{Z}_5$

Therefore, the set of integers divisible by 5 is countably infinite.

- (b) [6 Pts] $A \times \mathbb{Z}^+$ where $A = \{2, 3, 4\}$

Enumerate:

The Cartesian product $A \times \mathbb{Z}^+$ consists of all ordered pairs (a, n) where $a \in A$ and $n \in \mathbb{Z}^+$.

The enumeration can be written as: $(2, 1), (3, 1), (4, 1), (2, 2), (3, 2), (4, 2), (2, 3), (3, 3), (4, 3), \dots$

I can list all elements of A for each positive integer, moving upward in $\mathbb{Z}^+$.

Define Bijection:

Define: $f : \mathbb{N} \rightarrow A \times \mathbb{Z}^+$ by:

$$f(n) = (a_{n \bmod 3}, \lfloor \frac{n}{3} \rfloor + 1)$$

where: $a_0 = 2, \quad a_1 = 3, \quad a_2 = 4$

I have:

An enumeration of $A \times \mathbb{Z}^+$ and a function $f : \mathbb{N} \rightarrow A \times \mathbb{Z}^+$ that is a bijection

Therefore, the set $A \times \mathbb{Z}^+$ is countably infinite.

2. [10 Pts] Argue that the set of all finite strings over the alphabet Σ , where

$$\Sigma = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, /\},$$

is countable. Use this to argue that the set of positive rationals is countable.

Hint: Represent any positive rational as a finite string.

Let $\Sigma = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, /\}$ be a finite alphabet.

Let Σ^* denote the set of all finite strings over the alphabet Σ .

I can write: $\Sigma^* = \bigcup_{n=0}^{\infty} \Sigma^n$ where Σ^n is the set of all strings of length n .

Since Σ is finite, each Σ^n is finite: $|\Sigma^n| = 11^n$

Thus, Σ^* is a countable union of finite sets: $\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \dots$ and therefore, Σ^* is countable.

Every positive rational number $q \in \mathbb{Q}^+$ can be written uniquely as: $q = \frac{a}{b}$, where $a, b \in \mathbb{Z}^+$, $b \neq 0$

I can represent this rational number as a finite string: " a/b " $\in \Sigma^*$

This defines an injective function from $\mathbb{Q}^+$ to Σ^* : $f : \mathbb{Q}^+ \rightarrow \Sigma^*$, $f\left(\frac{a}{b}\right) = "a/b"$

Since Σ^* is countable, and $\mathbb{Q}^+$ injects into Σ^* , I can conclude that $\mathbb{Q}^+$ is countable.

3. [7 Pts] Prove that the set of functions from $\mathbb{N}$ to $\mathbb{N}$ is uncountable, by using a diagonalization argument.

Proof by Contradiction

Assume that the set of all functions from $\mathbb{N} \rightarrow \mathbb{N}$ is countable.

Then I could enumerate all such functions as a list:

$$f_0, f_1, f_2, f_3, \dots$$

Each f_i is a function from $\mathbb{N}$ to $\mathbb{N}$, and I can represent each function as an infinite sequence:

$$f_i = (f_i(0), f_i(1), f_i(2), \dots)$$

I can define a new function $g : \mathbb{N} \rightarrow \mathbb{N}$ by: $g(n) = f_n(n) + 1$

This function differs from each f_n at least at the n -th position.

Assume: $g = f_k$ for some $k \in \mathbb{N}$

Then, $g(k) = f_k(k)$

By definition of g , I have: $g(k) = f_k(k) + 1$

This implies:

$$f_k(k) + 1 = f_k(k) \text{ which is a contradiction.}$$

Therefore, the assumption that all functions $\mathbb{N} \rightarrow \mathbb{N}$ can be enumerated is false.

Hence, the set of all such functions is uncountable.

4. [9 Pts] Give an example of two uncountable sets A and B (along with a justification) such that $A - B$ is (a) finite (b) countably infinite (c) uncountably infinite

(a) Finite

Let $A = \mathbb{R}$, $B = \mathbb{R} \setminus \{0\}$

Then $A - B = \mathbb{R} - (\mathbb{R} \setminus \{0\}) = \{0\}$ which is a finite set.

Both A and B are uncountable, since removing a finite number of elements from $\mathbb{R}$ does not affect its cardinality.

(b) Countably Infinite

Let $A = \mathbb{R}$, $B = \mathbb{R} \setminus \mathbb{Q}$

Then $A - B = \mathbb{R} - (\mathbb{R} \setminus \mathbb{Q}) = \mathbb{Q}$ which is a countably infinite set.

Here, both A and B are uncountable, while $A - B = \mathbb{Q}$ is countable.

(c) Uncountably Infinite

Let $A = [0, 1]$, $B = [0, 1] \cap \mathbb{Q}$

Then $A - B = [0, 1] \setminus \mathbb{Q}$ which is the set of irrational numbers in $[0, 1]$, and hence uncountably infinite.

A is uncountable, B is countable, and removing countably many elements from an uncountable set still yields an uncountable set.

5. [12 Pts] Determine whether the following sets are countable or uncountable. Prove your answer. To prove countable, describe your enumeration precisely, using dovetailing. There is no need to define a bijection.

- (a) [6 Pts] the set of real numbers with decimal representation consisting of all 5's (5.5 and 55.555... are such numbers).

Explanation:

Each such number is made by placing a decimal point in a finite or infinite string of 5's.

For example: 5, 55, 0.5, 5.55, 0.555...

I can list all finite strings of 5's and insert the decimal point in every possible place.

Also, I can list infinite forms like 0.555..., 5.555..., etc.

I can enumerate all such numbers using dovetailing over all finite strings of 5's, all decimal point positions, and infinite tails like 0.555...

Since these can be listed one by one, the set is countable.

- (b) [6 Pts] the set of real numbers with decimal representation consisting of 5's and 7's.

Explanation:

I can represent any such real number as a string over the alphabet $\{5, 7\}$, with a decimal point at some position.

Consider the subset of such numbers in the interval $(0, 1)$, with infinite decimal expansions using only 5's and 7's, such as: 0.555..., 0.575575..., 0.777..., 0.757575...

Each such number corresponds to an infinite sequence over $\{5, 7\}$.

Define: $f : \{5, 7\}^{\mathbb{N}} \rightarrow (0, 1)$ where each sequence $(a_1, a_2, a_3, \dots)$ maps to the real number: $0.a_1a_2a_3\dots$

This function is injective, because different sequences produce different decimal expansions.

The set $\{5, 7\}^{\mathbb{N}}$ is uncountable.

Therefore, the set of real numbers whose decimal expansions consist of only 5's and 7's is also uncountable.

Counting

Suggested Reading: Rosen Sections 6.1 - 6.4.

These are the problems that you need to turn in. Always explain your answers and show your reasoning. **You will be graded based on how well you explain your answers. Just correct answers will not be enough!**

6. [4 Pts] How many integers between 1 and 60000, inclusive, are divisible by 3 or 4 or 5?

Inclusion-Exclusion Rule

Let:

$A = \{ \text{multiples of } 3 \}$

$B = \{ \text{multiples of } 4 \}$

$C = \{ \text{multiples of } 5 \}$

I need to find:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

$$|A| = \left\lfloor \frac{60000}{3} \right\rfloor = 20000$$

$$|B| = \left\lfloor \frac{60000}{4} \right\rfloor = 15000$$

$$|C| = \left\lfloor \frac{60000}{5} \right\rfloor = 12000$$

lcm = least common multiple

$$|A \cap B| = \left\lfloor \frac{60000}{\text{lcm}(3,4)} \right\rfloor = \left\lfloor \frac{60000}{12} \right\rfloor = 5000$$

$$|A \cap C| = \left\lfloor \frac{60000}{\text{lcm}(3,5)} \right\rfloor = \left\lfloor \frac{60000}{15} \right\rfloor = 4000$$

$$|B \cap C| = \left\lfloor \frac{60000}{\text{lcm}(4,5)} \right\rfloor = \left\lfloor \frac{60000}{20} \right\rfloor = 3000$$

$$|A \cap B \cap C| = \left\lfloor \frac{60000}{\text{lcm}(3,4,5)} \right\rfloor = \left\lfloor \frac{60000}{60} \right\rfloor = 1000$$

$$\begin{aligned} |A \cup B \cup C| &= 20000 + 15000 + 12000 - 5000 - 4000 - 3000 + 1000 \\ &= 47000 - 12000 + 1000 = 36000 \end{aligned}$$

Final Answer: 36000

7. [8 Pts] An ISU Computer Science shirt is sold in 8 colors, 6 sizes, striped or solid, and long sleeve or short sleeve. (a) How many different shirts are being sold? (b) What if the red and gold shirts only come in short-sleeve and solid?

(a) How many different shirts are being sold?

I am given 8 colors, 6 sizes, 2 patterns: striped or solid, and 2 sleeve types: long sleeve or short sleeve

$$\text{Total shirts} = 8 \times 6 \times 2 \times 2 = 192$$

192 shirts

(b) What if the red and gold shirts only come in short-sleeve and solid?

Assume: Red and gold (2 colors) are restricted to 1 sleeve type (short-sleeve) and 1 pattern (solid), and the other 6 colors are available in all combinations

$$\text{Count red and gold shirts: } 2 \times 6 \times 1 \times 1 = 12$$

$$\text{Count other shirts: } 6 \times 6 \times 2 \times 2 = 144$$

$$\text{Total shirts: } 12 + 144 = 156$$

156 shirts

8. [16 Pts] Let A and B be sets of 4 elements and 6 elements, respectively. How many different functions are possible from A to B ? from B to A ? How many different relations are possible from A to B ? How many of the functions from A to B are one-to-one? How many of the functions from B to A are onto?

(a) How many different functions are possible from A to B ? from B to A ?

Number of functions from A to B :

Each element of A can be mapped to any of the 6 elements in B .

Since there are 4 elements in A , the total number of functions is: $6^4 = 1296$

Number of functions from B to A :

Each element of B can be mapped to any of the 4 elements in A .

Since there are 6 elements in B , the total number of functions is: $4^6 = 4096$

(b) How many different relations are possible from A to B ?

A relation from A to B is a subset of the Cartesian product $A \times B$.

Since $|A| = 4$ and $|B| = 6$, I have $|A \times B| = 4 \times 6 = 24$.

The number of subsets of a set with 24 elements is: $2^{24} = 16777216$.

So, there are 16777216 different relations from A to B .

(c) How many of the functions from A to B are one-to-one?

A function from A to B is one-to-one (injective) if each element of A maps to a distinct element of B .

Since $|A| = 4$ and $|B| = 6$, I can count the number of injective functions as the number of ways to assign 4 distinct elements of B to the 4 elements of A :

Number of one-to-one functions = $P(6, 4) = \frac{6!}{(6-4)!} = \frac{6!}{2!} = \frac{720}{2} = 360$

(d) How many of the functions from B to A are onto?

A function $f : B \rightarrow A$ is onto if every element of A is used at least once in the image of f .

Since $|B| = 6$ and $|A| = 4$, I can compute the number of onto functions using the inclusion-exclusion principle.

Total functions = $4^6 = 4096$

Functions missing 1 value = $\binom{4}{1} \cdot 3^6 = 4 \cdot 729 = 2916$

Functions missing 2 values = $\binom{4}{2} \cdot 2^6 = 6 \cdot 64 = 384$

Functions missing 3 values = $\binom{4}{3} \cdot 1^6 = 4 \cdot 1 = 4$

Functions missing all 4 values = 0

Onto functions = $4096 - 2916 + 384 - 4 = 1560$

9. [12 Pts] A sack contains 50 movie tickets, 5 for each of 10 different movies (including 'Materialists'). Five friends want to go to a movie. (a) How many tickets would you have to remove from the sack to guarantee that everyone will be able to watch the same movie? (b) How many tickets would you have to remove from the sack to guarantee that everyone will be able to watch 'Materialists'? (c) State which problems above used the Pigeonhole Principle.

(a) How many tickets would you have to remove from the sack to guarantee that everyone will be able to watch the same movie?

To guarantee that 5 tickets for the same movie are drawn, I have to consider the worst-case scenario: draw 4 tickets for each of the 10 movies, totaling $4 \times 10 = 40$ tickets.

The next ticket (the 41st) must be the 5th ticket for some movie.

41 tickets

(b) How many tickets would you have to remove from the sack to guarantee that everyone will be able to watch 'Materialists'?

There are 5 'Materialists' tickets.

In the worst case, I might first draw all 45 tickets for the other 9 movies.

Only then can I be certain that the next 5 tickets are the ones for 'Materialists'.

50 tickets

(c) State which problems above used the Pigeonhole Principle.

The Pigeonhole Principle states that if more than kn items are placed into n containers, at least one container must hold more than k items.

In part (a), drawing 41 tickets ensures that some movie must have at least 5 tickets due to the Pigeonhole Principle.

In part (b), I am targeting a specific movie and exhausting the other possibilities, not applying the Pigeonhole Principle.

Therefore, only (a) uses the Pigeonhole Principle.

10. [6 Pts] In how many ways can a photographer arrange 8 people in a row from a family of 15 people, if (a) Mom and Dad are in the photo, (b) Mom and Dad are next to each other.

(a) Mom and Dad are in the photo

I must include Mom and Dad among the 8 people.

So I choose 6 additional people from the remaining 13 and then arrange all 8 people.

$$\binom{13}{6} \cdot 8! = (1716)(40320) = 69177600$$

(b) Mom and Dad are in the photo and next to each other

I treat Mom and Dad as a single unit.

I then choose 6 additional people from the remaining 13, then arrange the 7 units (Mom-Dad block + 6 others), and account for the internal arrangement of Mom and Dad within their unit:

$$\binom{13}{6} \cdot 7! \cdot 2 = (1716)(5040)(2) = 17297280$$

11. [4 Pts] A coin is flipped ten times where each flip comes up either heads or tails. How many possible outcomes contain at least four heads?

The total number of possible outcomes is:

$$2^{10} = 1024$$

I need outcomes that contain at least 4 heads, i.e., outcomes with 4, 5, ..., 10 heads.

The number of such outcomes is given by:

$$\sum_{k=4}^{10} \binom{10}{k} = \binom{10}{4} + \binom{10}{5} + \binom{10}{6} + \binom{10}{7} + \binom{10}{8} + \binom{10}{9} + \binom{10}{10}$$

Evaluating each term:

$$\binom{10}{4} = 210, \quad \binom{10}{5} = 252, \quad \binom{10}{6} = 210, \quad \binom{10}{7} = 120, \quad \binom{10}{8} = 45, \quad \binom{10}{9} = 10, \quad \binom{10}{10} = 1$$

Adding them together:

$$210 + 252 + 210 + 120 + 45 + 10 + 1 = 848$$

$$848$$

For more practice, you are encouraged to work on other problems in Rosen Sections 2.5 and in LLM Chapter 8.